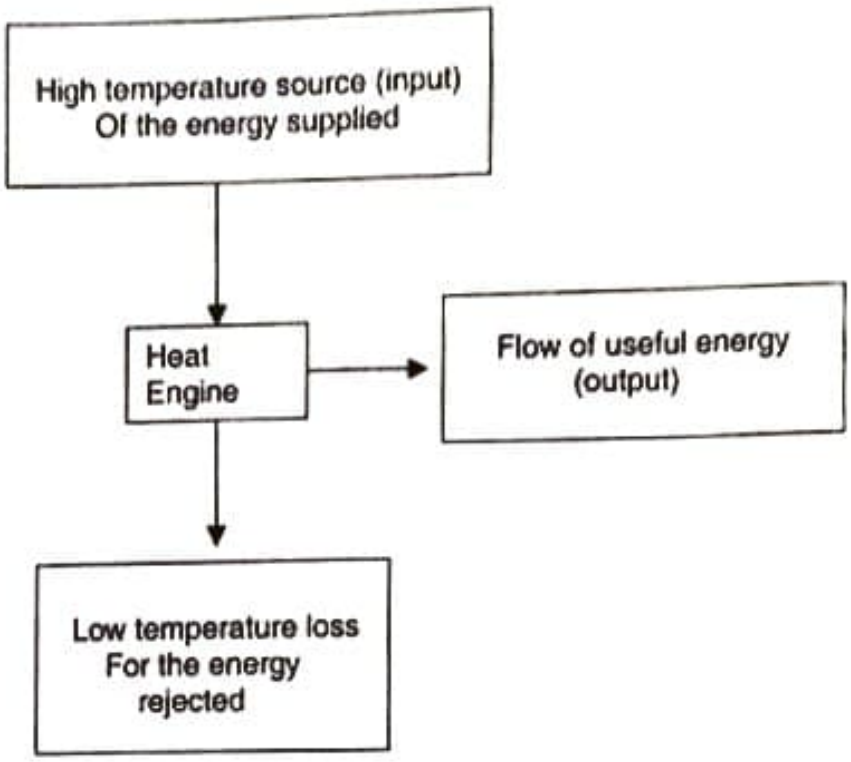
ICE

The function of a heat engine is to convert the energy, supplied to it During a transfer of heat energy, into some form of mechanical energy which can be used to drive, say, a propeller or an alternator.
The principle of such an engine is illustrated in the Fig. below.



The efficiency of the energy conversion system is defined thus;
Power plant efficiency = Flow of mechanical energy from plant (output)
Energy flow supplied to plant (input)

Definition of respective thermal efficiency.

1. **Theoretical thermal efficiency (Yth)**

Defined as the ratio of heat equivalent to theoretical work of the theoretical cycle to all heat supplied to the cycle.

$$\therefore \quad Yth = \frac{AW_{th}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = \frac{1 - Q_2}{Q_1}$$

A = thermal equivalent to mechanical work
W_{th} = theoretical work of a cycle
Q_1 = heat supplied
Q_2 = heat rejected

2. Indicated thermal efficiency (Yi)

The work done on the piston by combustion gases is called 'indicated work' and the power in this case is called 'indicated horsepower (IHP)' where the thermal efficiency in this case is called "Indicated thermal efficiency"

$$Yi = \frac{\text{Indicated work}}{\text{All heat supplied}} = \frac{AW_i}{Q_1}$$

W_i = indicated work
[Yi at its best efficiency equals Yth]

3. Brake (effective) thermal efficiency (Ye)

W_e, 'the effective work' which is actually gained from the engine, is expressed as; Effective work, W_e = indicated work - mechanical losses

The power which is actually gained from the shaft output is called, "Brake horsepower (BHP)" and the thermal efficiency calculated on the basis of effective work is called "effective(brake) thermal efficiency"

$$Ye = \frac{\text{effective work}}{\text{All heat supplied}} = \frac{AW_e}{Q_1}$$

. Mechanical efficiency (Ym)

Mechanical efficiency is given by;

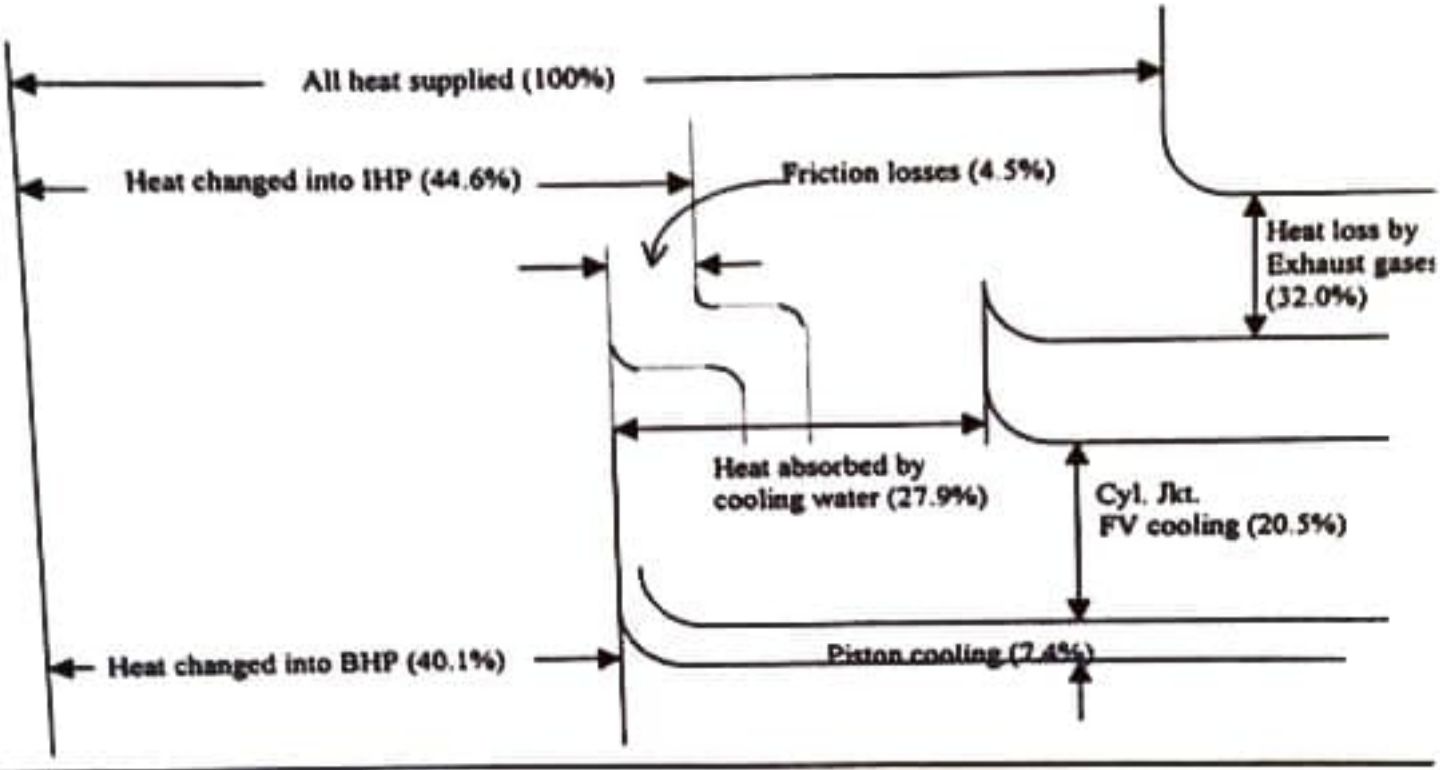
$$Ym = \frac{\text{effective work}}{\text{indicated work}} = \frac{W_e}{W_i} = \frac{BHP}{IHP} = ye / yi$$

The difference between the IHP and the BHP is equivalent to the work

Heat Balance.

Considering all the quantity of heat produced by perfect combustion of fuel oil injected into the cylinder be 100%, then. "Heat Balance" rates - how many percentages of all hat have been changed into "effective work" and how many percentages of it have been deprived by cooling water and exhaust gases, and how many percentages of it has been expended as friction losses respectively.

Therefore heat balance is said to show the aspect of heat distribution, which took place in the cylinder.
In general, heat balance may be acquired from the result of precise equipment and it also varies slightly with the change of engine load.
Following figure shows one example of a "heat balance" chart;



Theory of Combustion.

Combustion of fuel may be defined as;

1. A rapid chemical combination of two or more particular substance.
2. Oscillation of a chemical substance during which energy is released in a form of heat.

The form of combustion, which is made, used of for the production of power is the chemical combination of carbon and hydrogen of which liquid fuels are mainly composed of; with the oxygen contained in air. When the liquid fuels are heated beyond the combustion point; the carbon, hydrogen and sulphur contained in the fuels produces much heat when united with the oxygen contained in the air.

Quantity of heat produced by liquid fuels varies not only with the composition of fuel but also with the perfectness of combustion.

The three important aspect of combustion during the combustion process are;

a. Direction of reaction and how far it will go
b. Amount of energy released
c. Rate of which the reaction occurs

Air required for Combustion

To get complete combustion of fuel, at least, the minimum weight or volume of oxygen, namely air, is required, and needed to be supplied.
In order to get complete combustion actually, more additional quantity of air must be supplied.

Excess air factor is defined as the ratio of quantity of air actually supplied into the cylinder to the quantity of air theoretically required for perfect combustion.

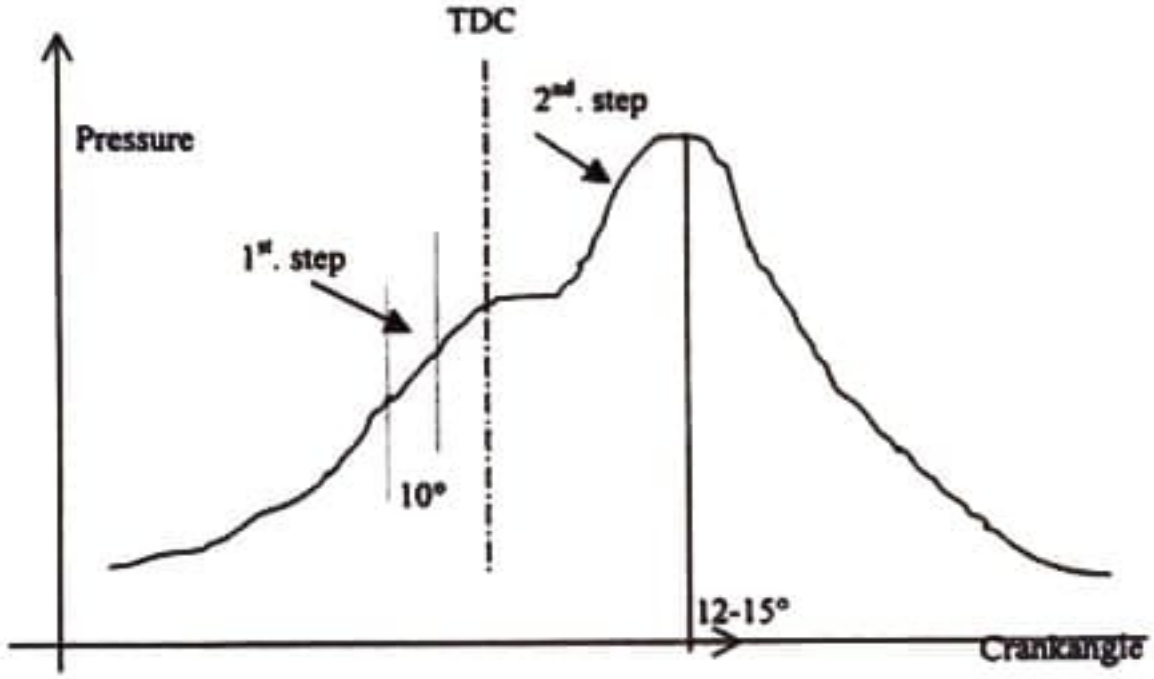
Limit of Combustion.

In actual cylinders, at first, where either spark-ignition or self-ignition takes place; for the mixed gases around one part to burn and the temperature rises thereby the heat produced being transferred to the adjoining combustibles by means of conduction, convection and radiation.

This heat transfer raises the temperature of the adjoining mixed gases to cause them to burn, resulting in the combustion of the mixed gases being propagated in sequence. However if the rate of mixture of air and fuel is too dense or too thin, the combustion cannot be propagated any further due to the temperature drop and thereby causing a failure in the burning of the mixed gases.

Hence, combustion limit is defined as a limit of mixed gases within which combustion can take place. The maximum limit is called the upper limit and the minimum limit is called the lower limit.

Combustion process in gasoline engine.



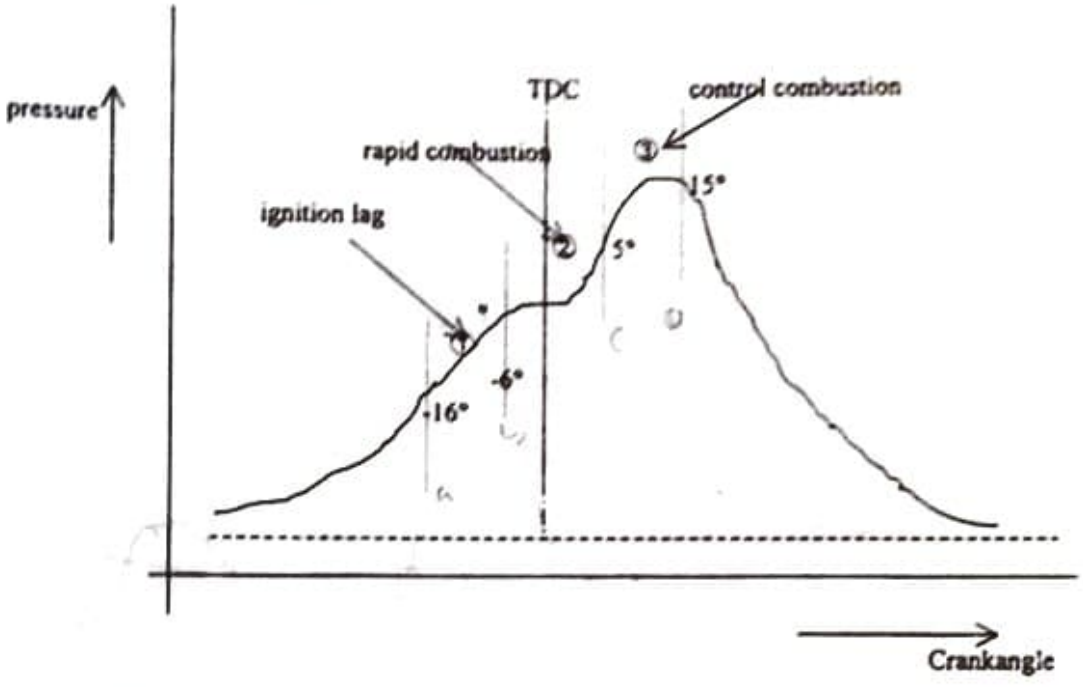
In general, the combustion process consists of the following two steps;

1.First step

Well vaporised combustible gases which are easily spark-ignited, are formed inside the cylinder at the end of compression stroke.

The first step is the period required for one part of combustible gaseous mixture to be ignited at first and then form a flame capable to propagate by itself. This period has much relation with the quality of fuels, mixing rate, temperature and pressure inside the cylinder. Furthermore there occurs no

1.First step: (period of ignition lag)
During this period, the injected fuel only absorbs heat from the hot air inside the cylinder, forming combustible gaseous mixture only. Thus there is no rapid pressure rise in this step. This period is also known as the ignition lag period and the length of this affects seriously the combustion of the gaseous mixture in the following step.



The following matters usually affect this period;

- quality of fuel oil
- pressure and temperature in the cylinder
- distribution and atomisation of the fuel oil.

ii. Second step: (period of rapid combustion)
At the end of the ignition lag, much combustible gaseous mixture which are heated to the vicinity of ignition temperature, had been formed everywhere in the cylinder.

Therefore when ignition once occur in one place or several places simultaneously, the combustion is propagated so rapidly that all combustible gaseous mixture are burned almost at once resulting in a rapid pressure rise in the cylinder.

The pressure rise in this case depends upon the equality of combustible mixed gases which have been formed during ignition lag period, the degree of atomisation of fuels and the ignition velocity of the liquid fuels.

iii. Third step: (period of controlled combustion)
The pressure and temperature inside the cylinder reaches to a very high state once the combustion has propagated and this causes the ignition lag of the injected fuel to be reduced remarkably. This phenomenon will bring about such a state that flame is injected from the injection nozzle.

In such a situation, the combustion inside the cylinder is then controlled by the rate of fuel injection. The pressure rise here is affected by the fuel injection rate, quantity of oxygen left inside the cylinder at the moment and the relative density between the fuel oil fragments and the hot air.

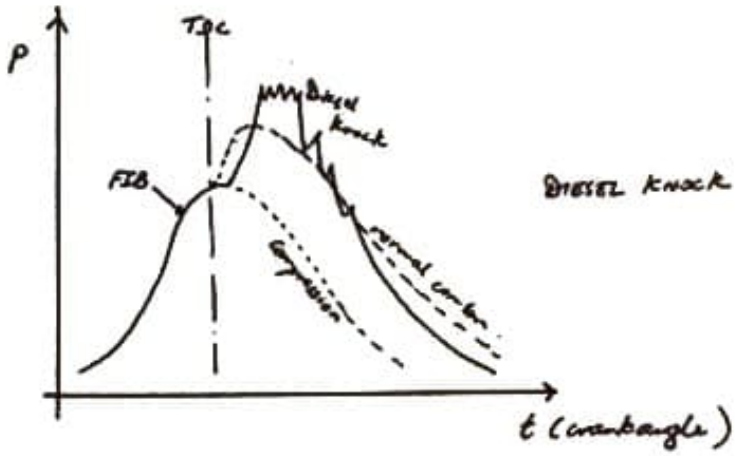
Ignition lag.
The period required for the liquid fuel to be ignited after the injection is usually defined as "ignition lag". This period consists of both a period of physical lag and a period of chemical lag.

The physical lag is the period required for the oil fragments to be vaporised with taking heat from hot air and to be heated to spontaneous ignition temperature, thus forming gaseous mixture.

The chemical lag is the period required for the mixed gases to be ignited actually after they have reached spontaneous ignition temperature.

Knocking.
"Gasoline knock" occurs only in spark ignition engines and at the end of

"Diesel knock" occurs at the beginning of combustion and is due to the pressure wave and mechanical variation caused by the combustion of mixed gases.
This knock is also caused by the longer ignition lag of liquid fuel.



When "diesel knock" had occurred, it does not have so much detrimental effects to the engine as in gasoline knock. However rapid combustion in the vicinity of the TPC makes the fuel consumption rate decreases and produces more power; but to avoid pressure rise and to keep calm operation of the engine, it is better that diesel knock should be avoided.

Methods to prevent the occurrence of both knocks can be seen from the comparison table below;

Remedial action on:	Gasoline engine	Diesel engine
Compression ratio	Low	High
Initial temp. of air supply	Low	High
Temp. of cylinder wall	Low	High
Revolution speed of engine	High	Low
Initial pressure of supply air	Low	High
Ignition point of fuel oil	High	Low
Ignition lag of fuel oil	Long	Short
Cylinder volume	Small	Large

Octane number is used in order to compare the degree of anti-knock property of each liquid fuel.

Cetane number is used to compare the degree of easiness of ignition of each liquid fuel.